CAFÉ-CD: Point-in-Time Causal Discovery on Factor-Adjusted Residuals

A short results paper — where factor residualisation wins, where it loses and why, what three public benchmarks are actually measuring, and what a causality certificate costs

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Abstract

CAFÉ-CD runs causal discovery on the idiosyncratic residuals of a strictly point-in-time dynamic factor model: CAFÉ residualisation, engine routing, backward-only lag screening, a panel group-lasso skeleton, and assumption-routed orientation with explicit abstention. It is numpy-only and CPU-only; every stage reads rows $\leq t$, so the graph reported at t is truncation-invariant by construction. This paper reports what we measured, including the parts that do not favour us. **(1) We win decisively on panel data.** On synthetic panels whose ground truth we audit ourselves, CAFÉ-CD is rank 1 of 15 at every common-variance share χ — 0.985 at χ=0.50 down to 0.929 at χ=0.95, against a best baseline that never leaves 0.708–0.720. A fairness arm re-runs every baseline on the *identical* CAFÉ residual and moves it by −0.048 to +0.086, so the margin is the discovery machinery, not the preprocessing. The baselines are single-series methods applied per entity: pooling the entity dimension is a real advantage and it is most of what the table measures. **(2) We lose on single-series benchmarks, and we measured why.** On the published CausalRivers cell CAFÉ-CD ranks 11 of 14, behind a 1 ms OLS VAR. Holding the estimator and the moment tensor fixed and varying only two design choices decomposes the loss: residualising costs 0.278 AUROC and conditioning on every other source rather than one costs a further 0.055. Part of that first term is a **located library defect**: CAFÉ's ARD rank selection is not scale-equivariant, so below unit scale the factor adjustment removes *nothing*, and on unrescaled river discharge it is mis-scaled rather than merely idle. We report it with its measured cost and its re-certification requirement, not as a pending win — no benchmark verdict changes. A second limit, measured on our own generator, is that the evidence blend's premium over its best single component is monotone in the number of entities and turns *negative* at E=1 — and every public single-series benchmark is E=1. We deliberately shipped no gate for it: the effect is a hundredth of an AUC and fitting one would be the benchmark-fitting we criticise. Three further hypotheses were tested and falsified, and are reported. **(3) Benchmark forensics.** We reproduce five published CausalRivers cells, then show the same VAR moves 0.849 → 0.593 on all 2 139 subgraphs by switching to a scale-free statistic; the true graphs have var-sortability 0.939. We reproduce the CausalDynamics `coupled` leaderboard from the authors' own shipped artefacts, and then find that a *constant identity matrix* scores 0.682/0.353 there and no shipped method beats it. Two `simple` cells are broken and we reproduce both published AUROCs to 10^{-3} by replicating the defect; a third defect feeds an endpoint matrix to a ranking metric. **(4) A point-in-time board**, now complete: every published method is batch, and churn and a leakage delta on each method's own metric say what that costs. **(5) Runtime**: 21× faster than PCMCI+ on a real 12-variable series at the same dense submission. Limitations get their own section.

1 The method

CAFÉ-CD is the discovery read-out of CAFÉ [1], a causal (strictly point-in-time) dynamic factor model whose forward pass already produces imputations, factors and residuals. The pipeline is five stages behind one public call, `cafe.discover(df)`:

1. **Residualise.** Take the factor-adjusted residual of the causal one-pass model. A latent common factor is a textbook confounder, so it is projected out by the same point-in-time pass that did the imputation. The residual is deliberately *not* winsorised: the tail is exactly what identifies edge direction in every LiNGAM-family branch downstream [15, 16].
2. **Route the loading engine** (panels only), §9.
3. **Screen lags** with a backward-only sign-correlation statistic whose threshold comes from a circular-block permutation null, not a constant. The encoding reads only indices $\leq t$; a forward-looking control is asserted to fail that test.
4. **Skeleton.** A panel group lasso on the surviving candidates fits a shared B_0 plus sparse per-entity deviations and selects the support.
5. **Orient.** Lagged edges are oriented by the arrow of time for free; only the contemporaneous ambiguous graph is

searched, per connected component. The orienter *abstains* — emitting `o-o` or `<->` — when no branch's assumptions hold, rather than manufacturing a DAG.

`mode="pit"` truncates the panel and runs the identical pipeline, so the reported graph is truncation-invariant by construction. Dependencies: numpy.

2 Scope, stated up front

The obvious objection to residualising before discovery is that it can only help if there is a factor worth removing. That is right, and the interesting question is where the boundary sits. We generate panels with a tunable common-variance share χ and feed three *identical* representations — raw, CAFÉ-CD residual, oracle residual (true factors removed) — to three downstream estimators. The sparse estimators are scored threshold-free by lambda entry order, so no model-selection rule enters anywhere. Table 1 and Figure 1 are the result.

The crossover depends on the estimator, and that is the point. Under an unpenalised OLS-VAR the crossover is $\chi^\star$=0.83; under the panel group lasso it drops to 0.68 and under a per-target L_1 VAR to 0.72. This is exactly the mechanism identified by the factor-adjusted network literature — FNETS [3] and Krampe [4] — made quantitative. A pervasive factor makes the *observed* VAR dense: every series loads on the factor, so every series Granger-causes every other. Density is not a problem for unpenalised OLS, which simply absorbs the factor because it is spanned by its own lagged regressors. It is fatal for a *sparse* estimator, whose entire premise is that the true transition matrix has few non-zeros. Removing the factor is therefore worth roughly 15 points of χ to a sparse estimator and nothing to a dense one. At χ=0.90 the group lasso gains $\Delta = +0.060$ with 8/8 paired wins and p=0.004; at χ=0.40 it *loses* 0.066.

Below the crossover the *representation* is worse than raw. There is no way to read Table 1 that hides this and we do not try. The routing rule is structural, not outcome-based (§9), so it cannot rescue a panel whose χ is low — and it should not: at low χ the estimated factor is mostly estimation error, and subtracting it destroys signal. **But this is a statement about the representation in isolation, not about the pipeline**, and §3 is where the two come apart: the full CAFÉ-CD pipeline wins at χ=0.50 by 0.265 AUC even though residualisation *alone* costs 0.056 there. Table 1 fixes the downstream estimator and varies the input; Table 2 varies the whole method. They are answering different questions and they do not contradict.

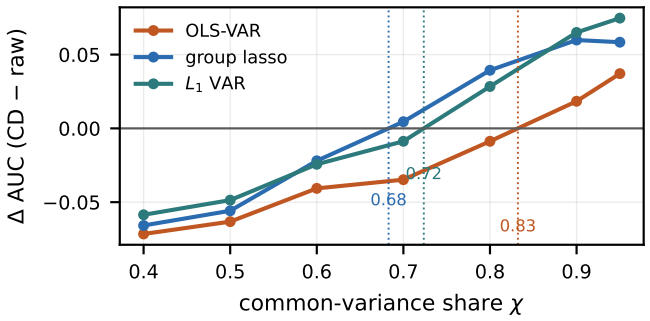


Figure 1: The crossover: paired advantage of the CAFÉ-CD residual over raw against χ, dotted lines at $\chi^\star$. The effect is a shallower decline, not a higher ceiling; Table 1 has the levels.

3 Head-to-head on ground truth we audit

The benchmarks in §4 turn out to measure things other than conditional-independence discovery, so the load-bearing comparison is the one where we control the data-generating process and can inspect the truth. We generate panels from `bench/gen_causal_panel.py` — E=10 entities, T=120, P=24, rank-3 common component, K=2 true lags, heterogeneous coefficients, Laplace shocks — sweep χ, and run 14 baseline configurations against CAFÉ-CD. Table 2 is the result: **rank 1 of 15 at every** χ, 0.985 at χ=0.50 falling only to 0.929 at χ=0.95, against a best baseline that never leaves 0.708–0.720.

Two things a reader would rightly ask, answered in the table itself. First, **is the margin just the preprocessing?** Every baseline is also run on the *identical* CAFÉ residual (the +RES column). If residualisation were doing the work, that arm would close the gap. It moves the baselines by −0.048 to +0.086 — mean +0.011 — against a gap of 0.219–0.267. It helps some and hurts others in the pattern §2 predicts rather than lifting everything: DYNOTEARS, already dense, is *worse* on the residual at every χ, while VAR-OLS crosses over and is better at high χ. The margin is the discovery machinery, not the input.

Second, **is the comparison fair?** Not in the sense of like-for-like architectures, and we would rather name that than let a reader find it. CAFÉ-CD is a panel method: it pools E entities into one shared skeleton with sparse per-entity deviations. The baselines are single-series methods; they are applied per entity and averaged, because that is what they can do. So a large part of what Table 2 measures is the value of the entity dimension. That *is* the contribution — a T=120, P=24 single series is hopeless and ten of them are not — but it is a claim about panels, and it does not transfer to the single-series benchmarks in §5, where it does not.

Table 1: **Factor residualisation helps only above a crossover in the common-variance share χ, and a *sparse* downstream estimator moves that crossover down.** Off-diagonal lagged-edge AUC for identical representations fed to three downstream estimators: RAW the observed panel, CD the CAFÉ idiosyncratic residual, ORC the oracle residual (true factors removed). The sparse estimators are scored threshold-free by lambda entry order, so no model-selection rule enters. Δ is the paired mean (CD − RAW) over 8 seeds, w the paired win count, p an exact one-sided sign-flip permutation test, $\chi^\star$ the interpolated crossover. Panel: E=10, T=60, P=24, n/col=12.1.

	OLS-VAR (unpenalised)				panel group lasso				L_1 VAR (per target)			
χ	raw	CD	orc	Δ w/p	raw	CD	orc	Δ w/p	raw	CD	orc	Δ w/p
0.40	0.979	0.907	0.992	-0.072 0/1.000	0.941	0.875	0.966	-0.066 0/1.000	0.982	0.923	0.995	-0.059 0/1.000
0.50	0.971	0.907	0.992	-0.063 0/1.000	0.922	0.866	0.966	-0.056 0/1.000	0.971	0.922	0.995	-0.049 0/1.000
0.60	0.958	0.917	0.992	-0.041 0/1.000	0.893	0.871	0.966	-0.022 1/0.996	0.951	0.927	0.995	-0.024 1/0.992
0.70	0.938	0.903	0.992	-0.035 0/1.000	0.852	**0.856**	0.966	+0.005 6/0.336	0.919	0.911	0.995	-0.009 2/0.727
0.80	0.906	0.897	0.992	-0.009 2/0.953	0.798	**0.838**	0.966	+0.039 8/0.004	0.871	**0.900**	0.995	+0.028 8/0.004
0.90	0.835	**0.854**	0.992	+0.018 7/0.008	0.717	**0.777**	0.966	+0.060 8/0.004	0.782	**0.847**	0.995	+0.065 8/0.004
0.95	0.753	**0.790**	0.992	+0.037 7/0.008	0.643	**0.701**	0.966	+0.058 8/0.004	0.692	**0.767**	0.995	+0.075 8/0.004
crossover	$\chi^\star = 0.83$				$\chi^\star = 0.68$				$\chi^\star = 0.72$			

Table 2: **Head-to-head on ground truth we audit ourselves: CAFÉ-CD is rank 1 of 15 at every χ, and the margin survives the fairness arm.** Off-diagonal lagged-edge AUC (`auc_cross`) on panels from `bench/gen_causal_panel.py` (E=10, T=120, P=24, K=2, Laplace shocks, heterogeneous β), 4 seeds per cell. **The baselines are single-series methods, applied per entity and averaged — so CAFÉ-CD's ability to pool E entities is a real advantage, and it is most of what this table measures.** The +RES column controls for the other half: each baseline re-run on the *identical* CAFÉ residual, which isolates the discovery machinery from the preprocessing. Across all 28 (method, χ) pairs residualisation moves a baseline by −0.048 to +0.086 (mean +0.011; it helps 17 of 28), so the gap to CAFÉ-CD is the machinery, not the representation. Runtime is mean seconds per cell on one CPU.

	$\chi = 0.50$		$\chi = 0.70$		$\chi = 0.90$		$\chi = 0.95$		
method	own	+RES	own	+RES	own	+RES	own	+RES	s/cell
CAFÉ-CD	**0.985**	—	**0.980**	—	**0.952**	—	**0.929**	—	4.01
DYNOTEARS	0.720	0.703	0.712	0.686	0.708	0.666	0.709	0.662	68.52
VAR-OLS	0.685	0.653	0.663	0.649	0.614	0.623	0.578	0.604	0.00
PCMCI	0.662	0.674	0.636	0.648	0.569	0.597	0.551	0.562	29.98
Granger (pairwise)	0.607	0.675	0.556	0.642	0.531	0.585	0.527	0.557	0.78
PC (lag-augmented)	0.614	0.624	0.559	0.588	0.513	0.555	0.500	0.541	122.91
DAGMA (linear)	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	17.35
VARLiNGAM	0.490	0.510	0.470	0.494	0.481	0.484	0.478	0.471	10.93
CAFÉ-CD rank	1/15		1/15		1/15		1/15		

4 Benchmark forensics

Before claiming anything on a public benchmark we reproduced it, and every criticism below is earned by a reproduction printed next to it. Our harness matches five published CausalRivers [5] cells across three graph sets and two metrics (Table 3a); DYNOTEARS — re-implemented from the paper [12], since `causalnex` does not install on Python 3.13 — and VARLiNGAM both land *above* the published figures.

The CausalRivers leaderboard is substantially varsortability. Discharge volume grows downstream, so on unrescaled data the raw VAR coefficient magnitude encodes edge direction directly. Table 3b holds the estimator, the data, the preprocessing and the sample fixed at all 2 139 flood/Random-5 subgraphs and changes only the scale convention. Scoring the same VAR with a scale-free t-statistic takes AUROC from 0.8493 to 0.5931. Rescaling in either of the two places a harness normally rescales — the benchmark's own min–max step, or a wrapper z-score — puts it *below* chance, at 0.4773 and 0.4256.

We then measured the prior rather than inferring it: over the same 2 139 true graphs, varsortability [8] is 0.939 — in 94% of true directed edges the effect has the larger marginal variance, so sorting nodes by variance *is* the causal order — and its scale-invariant counterpart R^2-sortability [9] is 0.376, so the order also leaks through predictability, in the opposite direction (Table 3b). This is the failure mode Reisach et al. [8] and Lohse & Wahl [10] found in *simulated* DAGs, here in a live ICLR 2025 benchmark on real data. It does not make CausalRivers a bad dataset — the scale prior is a genuine hydrological fact and a practitioner may use it — but the published 0.85 is not a measurement

Table 3: **(a) Our harness reproduces five published CausalRivers cells**, so (b) can be read as a statement about the benchmark rather than about our implementation. ‡DYNOTEARS is re-implemented from Pamfil et al. (2020) because `causalnex` does not install on Python 3.13; it and VARLiNGAM land *above* the published figures. **(b) The leaderboard is substantially var-sortability.** The identical VAR on all 2139 flood/Random-5 subgraphs, changing only the scale convention and the edge statistic — then the prior measured directly on the true graphs, with no model fitted. Varsortability is the fraction of true directed edges whose effect has the larger marginal variance (0.5 = no signal, 1 = sorting by variance *is* the causal order); R^2-sortability is its scale-invariant counterpart, so standardising does not remove it.

(a) reproduction of published cells

method	cell	metric	n	ours	pub.
VAR	flood/R-5, Fig. 3a	AUROC	2139	0.8493	0.85
VAR	east/R-3, Tab. 3	AUROC	1196	0.8229	0.82
VAR	east/R-3, Tab. 7	max-F_1	1196	0.8292	0.83
DYNOTEARS‡	flood/R-5	AUROC	2139	0.8157	0.79
VARLiNGAM	flood/R-5	AUROC	2139	0.7400	0.72

(b) four scale conventions, then the prior itself

scoring convention	scale prior	AUROC	max-F_1
$\lvert\hat{\beta}\rvert$, unrescaled *(published cell)*	✓	0.8493	0.6797
$\lvert t\rvert$, unrescaled *(scale-free)*	✗	0.5931	0.5154
$\lvert\hat{\beta}\rvert$, benchmark min–max rescale	✗	0.4773	0.4423
$\lvert\hat{\beta}\rvert$, wrapper z-score	✗	0.4256	0.4256
varsortability of the true graphs	—	**0.9393**	sd 0.110
R^2-sortability of the true graphs	—	0.3762	sd 0.254

of conditional-independence discovery, and a method that declines the prior is not being compared like for like. **The scale-free bar is** 0.593**.**

The prior, read method by method. Table 5 takes 150 of those subgraphs and asks what each method *gains* from being handed unrescaled rather than z-scored data. The field splits into exactly three groups, and the split is the published ranking. 5 methods rank edges by coefficient magnitude and gain +0.2448–+0.3085 AUROC; they are precisely the top of the published board. 7 rank by a test statistic or a conditional-independence test and are invariant to four decimal places — Granger, PC, FCI, PCMCI and PCMCI+ move by exactly 0.0000. And CAFÉ-CD alone is *harmed*, by −0.1012; §6 shows that part of that is a library defect of ours rather than a property of the estimator. A leaderboard whose ordering is this cleanly predicted by a preprocessing convention is not primarily measuring discovery.

Two CausalDynamics `simple` cells are broken. On the `simple` split of CausalDynamics [6] we reproduce two published AUROCs to 10^{-3} — but only by replicating their defects (Table 6). Their DYNOTEARS wrapper builds a cause-row matrix and scores it against effect-row ground truth, which is precisely why the published 0.426 sits *below* chance; correcting the orientation moves it to 0.489, from "clearly wrong" to "chance". Their VARLiNGAM wrapper binarises an unpruned VAR-LiNGAM lag matrix, which is dense, so the prediction is all-ones and 0.500 is true by construction rather than by measurement — as it is for `cutsplus` and `rcd`, both also exactly 0.500. More broadly, on this split nothing beats chance for anyone, so it is a fine stress test but not a place where a win means anything today.

On `coupled`, a constant beats the whole leaderboard. The `coupled` tier is the one the benchmark leads with, and it is not degenerate the way `simple` is: four of the ten shipped methods are above 0.5 on 96–100% of the 100 Default-row systems individually. We re-derived that row from the authors' own shipped `outputs/coupled.tar.gz` — five of seven published cells land within 0.026, so the row is reproduced (Table 7a) — and only then scored a **constant identity matrix** through the same evaluator. It gets 0.682 AUROC / 0.353 AUPRC, and paired system by system (Wilcoxon signed-rank, n=100) **not one of the nine testable methods beats it**. PCMCI+, the best of them, ties on AUROC (p=0.21) and is significantly *worse* on AUPRC (p=5×10^{-18}), winning on 2% of systems. The mechanism is arithmetic: the protocol keeps the diagonal, the coupled ground truth's diagonal is 63% dense against a 10% dense off-diagonal, so 40.8% **of the entire positive class is self-loops**. Anything that fires on autocorrelation — trivial in an ODE system — banks 41% of the positives before looking at a single cross-variable relation. On `simple` the same predictor scores 0.453, below chance, so this is specific to `coupled`.

The published AUPRC null is too low on both tiers. The benchmark's own `Null` comes from an *all-zeros* prediction, for which average precision collapses to the positive rate. Against a uniform-random *ranker* — 20 draws per system through the benchmark's own scorer — the AUPRC floor is 0.188 on `coupled` (theirs: 0.153) and 0.759 on `simple` (theirs: 0.687). **Every published `simple` AUPRC, .67–.70, is below chance**; on `coupled`, NGC and CUTS+ are below it and VARLiNGAM is at it.

Where the tier *is* a real venue, and a third defect. Two of the five published `coupled` rows are usable and we say which: on **Confounder** the diagonal carries only 19.0% of the positives and on **Lag** 17.0%, the identity floor falls to ≈ 0.545, and methods separate from it (Table 7c). The sobering part is what clears the bar: a plain **OLS VAR(1) at ∼1 ms**, ranked by $|t|$, scores 0.590/0.312 on Confounder against PCMCI+'s 0.593/0.208 — level on AUROC, half again as good on AUPRC — and ranked by $|\hat{\beta}|$ scores 0.607/0.300 on Lag against 0.560/0.236, clearing it on both. So the bar on the rows worth competing on is a linear VAR, not the leaderboard. Two corrections belong with this. First, the transposition defect is real on `coupled` too but hits the *other* half of the suite, because the two subsets store opposite conventions: DYNOTEARS is correctly

(a) The engine router: a structural test with a null of 1. *within* is the alignment of an entity's own two half-sample loading subspaces — the ceiling identical loadings would give; *between* is the same alignment across entities. The raw *between* column interleaves across the two families (0.840–0.936 homogeneous vs. 0.857–0.942 heterogeneous), so no absolute cutoff orders them; the ratio does. MAE is held-out (10% MCAR) with each engine *pinned* — an outcome the router never sees. It picks the lower-MAE engine on 8/8 panels.

		router (design matrix only)				held-out MAE		
panel	seed	within	betw.	ratio	picks	pooled	per-e.	ok
homog.	0	0.850	0.840	0.988	pooled	1.648	1.875	✓
	1	0.941	0.936	0.995	pooled	1.882	1.936	✓
	2	0.907	0.908	1.001	pooled	1.826	2.007	✓
	3	0.877	0.892	1.017	pooled	1.854	2.064	✓
heterog.	0	0.999	0.857	0.858	per-e.	0.386	0.196	✓
	1	1.000	0.942	0.943	per-e.	0.435	0.231	✓
	2	0.999	0.892	0.893	per-e.	0.496	0.265	✓
	3	1.000	0.928	0.928	per-e.	0.510	0.244	✓

(b) A null result on our own idea: bit-packed lag screening is 8.1–21.4× *slower* than dense BLAS, with no crossover. Statistic only (n_{perm}=0), n=4000, K=8, best of 3, single process, macOS-arm64, numpy 2.5.0. The two backends return *bit-identical* statistic surfaces ($\max|\mathrm{d}S| = 0$ in every cell), so this is a pure speed comparison, and the gap *widens* with P. At 20% MCAR the ordering is unchanged (15.2× at P=100).

	`stat="phi"`			`stat="arrow"`		
P	dense (s)	bits (s)	ratio	dense (s)	bits (s)	ratio
20	0.0015	0.0119	8.1×	0.0016	0.0167	10.6×
50	0.0075	0.0961	12.8×	0.0130	0.1647	12.7×
100	0.0221	0.3404	15.4×	0.0271	0.5637	20.8×
200	0.0808	1.3474	16.7×	0.1035	2.2151	21.4×

Table 4: **Two engineering results from building CAFÉ-CD, one of them negative.** Both are live re-runs, not transcriptions: `scripts/run_engine_router.py` and `scripts/run_screen_backend.py`.

Table 5: **The CausalRivers leaderboard, decomposed per method: what each one gains from being handed unrescaled rather than z-scored data.** Same 150 flood/Random-5 subgraphs, same estimator, same lag budget; only the scale convention differs. The field splits into exactly three groups and the split is the published ranking: the 5 methods that rank by coefficient magnitude gain 0.24–0.31 AUROC and are precisely the top of the published board; the 7 that rank by a test statistic or a conditional-independence test are invariant to 4 decimal places; and CAFÉ-CD alone is *harmed*. This is Table 3b's var-sortability prior (0.939 over 2 139 subgraphs) read method by method.

method	unrescaled	z-scored	gain
coefficient magnitude — gains			
DYNOTEARS	0.7663	0.4578	+0.3085
DAGMA (linear)	0.8231	0.5356	+0.2874
VAR-OLS	0.8490	0.5955	+0.2534
VARLiNGAM	0.7427	0.4972	+0.2455
DirectLiNGAM	0.7420	0.4972	+0.2448
test statistic / CI test — exactly invariant			
FCI	0.5108	0.5108	+0.0000
FCI (lag-augmented)	0.4893	0.4893	+0.0000
Granger (pairwise)	0.7089	0.7089	+0.0000
PC	0.5105	0.5105	+0.0000
PC (lag-augmented)	0.5151	0.5151	+0.0000
PCMCI	0.5948	0.5948	+0.0000
PCMCI+	0.5324	0.5324	+0.0000
scale-invariant by construction — harmed			
CAFÉ-CD	0.4524	0.5536	**-0.1012**
cafe_cd_nocontemp	0.4260	0.5443	**-0.1183**

Table 6: **Two CausalDynamics cells are wrong by construction; we reproduce both published AUROCs to 10^{-3} by replicating the defect.** `simple`, `noise=0.00_confounder=False`, 54 scored systems, $\tau_{\max}$=1, self-loops scored, mean per-realisation. [a] scores a cause-row matrix against effect-row ground truth — which is why its published AUROC is *below* chance; correcting the orientation moves it to chance. [b] binarises an unpruned, dense lag matrix, so the prediction is all-ones and 0.500 is an identity, not a measurement. Across the ten methods the benchmark ships for this setting, AUROC spans 0.426–0.518 against a 0.500 null and no method is more than 0.020 AUPRC from the 0.687 constant-prediction baseline: nothing beats chance here, for anyone.

	ours		published	
variant	AUROC	AUPRC	AUROC	AUPRC
DYNOTEARS, upstream[a]	0.4251	0.6750	0.4259	0.6752
orientation fixed	0.4892	0.6954	–	–
VARLiNGAM, upstream[b]	0.5000	0.6872	0.5000	0.6927
`lingam` defaults	0.5149	0.6992	–	–
plain VAR, $\lvert\hat{\beta}\rvert$	0.5245	0.7265	–	–

oriented there and VARLiNGAM and RCD are transposed — so DYNOTEARS' published coupled .66 is not affected by the bug that produced its below-chance .43 on `simple`. Second, a defect not previously recorded: `grasp`'s wrapper feeds causal-learn's **endpoint** matrix ($-1/0/+1$) straight into `roc_auc_score` *as a score*, so a -1 tail endpoint ranks *below* the 0 that means "no edge". Measured on ten coupled systems, 18.6% of the submitted matrix is -1 and taking `abs()` first moves AUROC from 0.442 to 0.545: the encoding alone costs 0.103.

5 Where CAFÉ-CD loses

The same harness that earns the criticisms above scores us, and it is not kind. Table 8 is the head-to-head on both public single-series benchmarks.

On the published CausalRivers cell — flood/Random-5, unrescaled, the exact protocol that produces the 0.85 VAR number — CAFÉ-CD is 11**th of** 14 on AUPRC, at 0.257 against DYNOTEARS's 0.625 and VAR-OLS' 0.614. Mov-

Table 7: **CausalDynamics `coupled`: reproduced first (a), then a constant beats all of it (b).** **(a)** re-derived from the authors' own shipped `outputs/coupled.tar.gz` over the Default row's 100 systems — five of seven published cells land within 0.026, so the row is reproduced. *const* is the share of cells whose AUROC is *exactly* 0.500, i.e. a constant prediction; p and *wins* are a paired Wilcoxon signed-rank test of that method's AUPRC against a **constant identity matrix** over the same systems. **Not one of the nine testable methods beats the identity matrix**; `pcmciplus` merely ties on AUROC (p=0.21) and loses on AUPRC. **(b)** label-free predictors pushed through the benchmark's own scorer, 20 draws per system. Their shipped `Null` is an *all-zeros* predictor, which is the wrong reference for a ranking metric: against a random *ranker* the AUPRC floor is 0.188 on `coupled` and 0.759 on `simple`, so every published `simple` AUPRC (.67–.70) is below chance. **(c)** the mechanism is the diagonal: the protocol keeps it, and where it carries less of the positive class the identity floor falls and methods separate — but the bar there is a plain OLS VAR(1) at ∼1 ms, not the leaderboard.

(a) the shipped Default row, re-derived — and paired against a constant

		ours (shipped eval)		published		vs. identity (AUPRC)	
method	n	AUROC	AUPRC	AUROC/AUPRC	const	p	wins
`pcmciplus`	100	0.676	0.264	0.68 / 0.27	0%	5e-18	2%
`tsci`	100	0.654	0.323	0.69 / 0.36	0%	4e-04	33%
`fpcmci`	100	0.646	0.241	0.67 / 0.24	23%	5e-18	2%
`dynotears`	100	0.621	0.259	0.66 / 0.32	0%	8e-17	9%
`varlingam`	100	0.588	0.187	0.57 / 0.18	4%	4e-18	0%
`cutsplus`	100	0.504	0.159	0.50 / 0.16	56%	4e-18	0%
`ngc_lstm`	100	0.500	0.153	0.50 / 0.16	0%	4e-18	0%
`rcd`	100	0.500	0.155	—	28%	4e-18	0%
`tcdf`	3	0.497	0.156	—	0%	–	–
`grasp`	31	0.489	0.155	—	3%	9e-10	0%
constant identity	100	**0.682**	**0.353**	—	—	—	—

(b) the floor, measured

		as scored		off-diagonal	
tier	predictor	AUROC	AUPRC	AUROC	AUPRC
coupled	random ranker	0.499	0.188	0.495	0.140
	all-zeros (`Null`)	0.500	0.153	0.500	0.100
	identity matrix	0.682	0.353	0.500	0.100
simple	random ranker	0.493	0.759	0.500	0.806
	all-zeros (`Null`)	0.500	0.687	0.500	0.723
	identity matrix	0.453	0.681	0.500	0.723

(c) where the tier is a real venue (AUROC/AUPRC)

					OLS VAR(1), ∼1 ms	
row	n	diag.	ident.	`pcmciplus`	by $\lvert\hat{\beta}\rvert$	by $\lvert t\rvert$
Confounder	101	19%	0.546/0.183	0.593/0.208	0.543/0.244	0.590/0.312
Lag	103	17%	0.544/0.236	0.560/0.236	0.607/0.300	0.545/0.306

Table 8: **Where CAFÉ-CD loses: both public single-series benchmarks.** **(a)** CausalRivers head-to-head, every method the harness has finished on the identical subgraph sample, ranked by *individual* AUPRC (the metric to read on graphs this sparse); the *rank* column prints its own denominator. CAFÉ-CD is beaten by `var_ols` — a plain OLS VAR at ∼1 ms — in 5 of the 5 completed configurations. **(b)** CausalDynamics `simple`, `noise=0.00_confounder=False`, 54 systems scored by all 13 methods, CAFÉ-CD re-run live with the current library (`scripts/run_dynamics_matched.py`); ✓ marks the three entries above the *measured* random-ranker AUPRC floor of 0.759 for that tier (§4) — ten of thirteen, CAFÉ-CD included, are below it. Neither loss is ambiguous and §6 measures the cause.

(a) CausalRivers, individual AUPRC

			CAFÉ-CD			best method		`var_ols`
config	arm	n	AUPRC	AUROC	rank	method	AUPRC	AUPRC
east/conf-3	scale-exploiting *(published)*	24	0.565	0.529	**6/13**	DirectLiNGAM	0.731	0.696
east/R-3	scale-exploiting *(published)*	150	0.549	0.547	**8/13**	VAR-OLS	0.792	0.792
east/root-3	scale-exploiting *(published)*	150	0.613	0.641	**8/13**	DAGMA (linear)	0.741	0.739
flood/R-5	scale-exploiting *(published)*	150	0.257	0.452	**11/14**	DYNOTEARS	0.625	0.614
flood/R-5	scale-free	150	0.345	0.554	**4/14**	Granger (pairwise)	0.472	0.416

(b) CausalDynamics `simple`, 54 matched systems, AUPRC — 1. PCMCI+ 0.800 ✓; 2. PCMCI 0.774 ✓; 3. VAR-OLS 0.771 ✓; 4. VARLiNGAM 0.757; 5. Granger (pairwise) 0.747; 6. **CAFÉ-CD** 0.729; 7. DirectLiNGAM 0.725; 8. DAGMA (linear) 0.713; 9. DYNOTEARS 0.705; 10. FCI 0.700; 11. PC 0.698; 12. FCI (lag-augmented) 0.689; 13. PC (lag-augmented) 0.689. CAFÉ-CD is 6 of 13.

ing to the scale-free arm, where the var-sortability prior is removed and every method's score falls, lifts us to 4th of 14 at 0.345 — still behind Granger (pairwise)'s 0.472. **Part of the gap between our own two arms is a located library defect rather than inference quality**: on unrescaled input CAFÉ's factor adjustment is mis-scaled and the method's central mechanism is not doing what it claims (§6, Table 10). On the `east` configurations we place mid-

table and never first (Table 8a). **CAFÉ-CD is beaten by `var_ols` — an OLS VAR at ∼1 ms — in all 5 completed configurations**, and that is the fact to sit with rather than the arm in which we place best.

On CausalDynamics `simple` we are 6th of 13 over the 54 systems every method scored, at 0.729 AUPRC against PCMCI+'s 0.800. That tier's measured random-ranker floor is 0.759 (§4), so 10 of the 13 entries including ours are *below chance* and the ranking there means little either way. On `coupled` and `climate` the two boards overlap on only 7 and 11 fully-scored systems — the baseline board covers 10 of the 132 coupled systems CAFÉ-CD has run — which is too few to rank, so no number is drawn from them.

6 Diagnosis: which design choice costs the ranking

A loss is only useful if it localises. CAFÉ-CD differs from the baselines it loses to in exactly two ways, and both can be varied while holding the estimator and the moment tensor fixed: the *representation* (raw panel vs CAFÉ residual) and the *conditioning set* (for target i and source j: i's own past plus *all* other sources, which is what the skeleton does, or plus exactly *one* other, which is bivariate Granger). All four cells are the same $|t|$ statistic computed from the same moment tensor by `cafe.discovery._refit_t` and `cafe._skeleton.bivariate_granger`, so a difference between two rows is that one design choice and nothing else. Table 9 runs the 2×2 over 40 flood/Random-5 subgraphs.

The decomposition is unambiguous, and the identity checks earn it: the raw-conditional row reproduces `var_ols` and the raw-bivariate row reproduces `granger_pairwise` to 0.0000 mean absolute AUROC, on all 40 samples. Then:

1. **The representation costs** 0.278 **AUROC** — 0.771 → 0.493 at fixed conditioning, with 38/40 samples favouring raw. This is the dominant term by a wide margin (see the caption of Table 9 for the ratio).
2. **The conditioning set costs a further** 0.055 — 0.493 → 0.548 by dropping to one competitor at a time (23/40 paired samples), and 0.077 on the raw panel (34/40). On the raw panel this is not in doubt: it reproduces *exactly* the gap between `var_ols` and `granger_pairwise`, which is why pairwise Granger is the best method in the scale-free arm. **On the residual it does not replicate.** A separate 40-sample measurement taken while building the scoring router found the ordering *reversed* there — bivariate 0.578 against conditional 0.660 — under a different conditioning protocol. Enlarging our own sample from 30 to 40 shrank the gap without flipping it, so the disagreement is protocol, not sampling. Two measurements of the same contrast disagree, so we bank nothing from it: the bivariate ceiling is a property of the *raw* panel and we do not claim it survives residualisation. The shipped `_scoring_router` accordingly abstains on our own panels — the maximum change it makes to any of the 15 cached head-to-head cells is $|\Delta|$=0.0007 — and no gate was built on this.
3. **Three explanations we ruled out.** It is not that our ranking is uncalibrated but the support is fine: sweeping the threshold peaks at max-F_1 0.443, no better than the AUROC suggests. It is not marginal-vs-conditional in general: a naive marginal lagged |corr| with no conditioning at all scores 0.616, which beats our whole pipeline's 0.541 but loses to both raw rows. And it is not a router failure — the scoring router *does* fire here, measuring source tolerance 0.46 — a generalised variance-inflation reciprocal in the sense of Fox & Monette [11] — against a null of 1, and blending in w_{biv}=0.40 of the bivariate branch, which is why CAFÉ-CD end-to-end (0.541) sits between the two residual rows. It cannot recover a loss whose larger half is upstream of it.

Why the representation costs so much here: a located defect. The panels are five discharge stations on one river network — mean off-diagonal |corr| 0.746, max 0.932, condition number 63.9, leading eigenvalue 0.800 of the total — so a confounding common component is exactly what CAFÉ is built to remove, and the −0.278 above is the opposite of what the design predicts. It is worth asking whether the factor adjustment is doing what the method says it does. It is not.

Table 10 varies one global multiplier on a panel of our own and measures the variance share the factor adjustment actually removes, from one fit, so every non-factor term cancels. A factor model must be equivariant under a global rescale — rescaling the data rescales the loadings and nothing else. **CAFÉ is not.** The adjustment removes 74% of the variance at unit scale, 1.45% at a mean standard deviation of 0.375, and **exactly nothing** below that. The cause is one line: `src/cafe/_core.py` sets the ARD precision $\alpha_l = N/(\|W_{\cdot l}\|^2 + 1)$, and under $x \to cx$ the loadings scale as $W \to cW$ while the $+1$ does not. Below unit scale it dominates, every column's precision goes to its maximum, ARD kills every factor, and the "factor-adjusted" residual has no factor removed from it.

CausalRivers sits astride that floor. Within a single five-station panel the column standard deviations span 0.030 to 175.5 and *every* sample of the 25 we measured has at least one sub-unit column, so the floor bites on part of the panel and not the rest. The result is worse than idleness: unrescaled, the share removed is **negative on** 14 **of** 25 **samples** — the adjustment *adds* idiosyncratic variance, subtracting a component fitted to the columns that survived ARD from the columns that did not. z-scoring puts every column on the same side of the floor and the share becomes small but mostly positive (0.041, negative on 4 of 25). **That is the same pair of arms as the benchmark result**: unrescaled 0.4524 AUROC, z-scored 0.5536.

A trap, and a hypothesis this falsifies. Two things follow that we would otherwise have got wrong. First, **`effective_rank` is not a usable indicator of whether a factor was removed**: it reads 0 in every row of Table 10, including the rows removing 74% of the variance. Every

Table 9: **The loss decomposed: hold the estimator fixed, vary only the representation and the conditioning set.** 40 CausalRivers flood/Random-5 subgraphs (the published cell; $T{\approx}7025$, $P{=}5$, 4.0 true off-diagonal edges, $K{=}3$), off-diagonal, mean over samples. Every row is the *same* $|t|$ statistic from the *same* moment tensor — `cafe.discovery._refit_t` for the conditional cells, `cafe._skeleton.bivariate_granger` for the bivariate ones — so a row difference is that one design choice and nothing else. The *reproduces* column is the identity check: the two raw rows match two external baselines to 0.0000 mean absolute AUROC, on all 40 samples. w is the paired count of samples on which the row beats the one above it. **Both of CAFÉ-CD's choices cost on this benchmark, and the representation costs 5× more**: residualising is −0.278 AUROC at fixed conditioning (38/40 samples favour raw), and conditioning on every other source rather than one is a further −0.055 (−0.077 on the raw panel). The panel is near-collinear — mean off-diagonal |corr| = 0.746, max 0.932, condition number 63.9, leading eigenvalue 0.800 of the total — which is the regime in which a fully conditional coefficient stops being readable, and the source-separability diagnostic the scoring router reads agrees ($\overline{\text{tol}} = 0.46$ against a null of 1, so it blends in $w_{\text{biv}} = 0.40$ of the bivariate branch). A confounding common component is exactly what CAFÉ is built to remove, so the sign of the first row difference is the opposite of what the design predicts. Table 10 measures why: on unrescaled input of this magnitude the factor adjustment is *mis-scaled*, so the CAFÉ-CD residual row here is not a clean confounder removal. A real gap survives that defect — see §6.

representation	conditioning set	AUROC	AUPRC	max-F_1	w	reproduces
raw panel	own past + **all** others	0.771	0.653	0.647	—	`var_ols`
raw panel	own past + **one** other	0.848	0.672	0.684	34/40	`granger_pairwise`
CAFÉ residual	own past + **all** others	0.493	0.275	0.429	—	*CAFÉ-CD's skeleton*
CAFÉ residual	own past + **one** other	0.548	0.299	0.460	23/40	
CAFÉ-CD, end to end		0.541	0.313	0.443	—	
raw marginal lagged |corr|, *no* conditioning		0.616	0.504	0.520	—	

statement in this paper about whether the factor engaged is made from the adjusted-versus-kept residual comparison instead. (On the CausalRivers samples the same accessor reports a mean rank of 3.5 while the adjustment is removing a *negative* share — the diagnostic and the measurement disagree in both directions.)

Second, the obvious objection — "residualisation removes the scale prior, and that is why CAFÉ-CD loses the scale-exploiting arm" — is **falsified**. Over the same 25 subgraphs, varsortability is 0.99 on the raw series and 0.76 on the CAFÉ residual, still far above the 0.5 no-signal point (R^2-sortability 0.30 against 0.33). The variance ordering largely survives residualisation. The correct statement is that the information is still in the residual and **our estimator is built not to read it**: every component of the score — the refit $|t|$, the path occupancy, the screening sign-statistic — is scale-invariant by construction. That is precisely why CAFÉ-CD shares the exactly-0.0000 signature with Granger, PC, FCI and PCMCI in Table 5, and it is a design choice we would make again.

What we are not claiming. The defect is located, reproducible and costly, and it is still not a pending win. Correcting it would move the scale arm towards the free-arm number; it would not reach `var_ols`' 0.8490 there, and **no verdict in Table 8 changes**. Nor is the fix free. The offending line is in the production imputation path, so changing it changes `impute()` for every small-scale dataset and requires re-running the main CAFÉ benchmarks to their 10^{-12} tolerance. Standardising inside `discover()` instead is not free either: `mode="pit"` truncates and then runs the identical pipeline, so a scale estimated from the data would become prefix-dependent and could break the REV=0 certificate of §7. We report a located defect with a measured cost and a re-certification requirement, and we have shipped neither route.

What remains after the defect. Removing it does not dissolve the −0.278. On the z-scored arm, where the mechanism does engage, CAFÉ-CD still scores 0.5536 against `granger_pairwise`'s 0.7089, so a real gap remains and the conditioning-set contrast of Table 9 still applies to it. What the defect changes is the attribution: the difference between our own two arms is a library bug, and only what is left after that is a statement about discovery on near-collinear single-series data. A router at the skeleton stage, on the same structural discipline as the scoring router, is still the obvious next step — but the first thing to fix is upstream of it.

A second limit, measured on our own generator: the evidence blend needs cross-entity replication. The result above is measured on a benchmark whose properties we have just spent two sections questioning, so the design question behind it is worth asking where we control the truth. CAFÉ-CD's edge evidence is a rank-blend of three statistics: the conditional refit $|t|$, the group lasso's path occupancy, and the screening statistic. Table 11 varies *only* the number of entities E and reads the *premium* — the blend's AUC minus that of its strongest single component — with ET held constant, so the contrast is entities and not rows.

The premium is **monotone in E and crosses zero be-**

Table 10: **CAFÉ's factor adjustment is not scale-equivariant, and `effective_rank` does not report it.** One panel (E=10, T=120, P=24, χ=0.90), one global multiplier, everything else identical — a global rescale of the data rescales the loadings and should change nothing. *removes* is the per-column variance share the factor adjustment actually takes out, read from ONE fit as `residuals(factor_adjusted=True)` against `(=False)`, so level, season, fixed effects and AR cancel exactly. The mechanism is off below unit scale and degrades again far above it; the cause is `src/cafe/_core.py`, where the ARD precision is $\alpha_l = N/(\|W_{\cdot l}\|^2+1)$ and the +1 does not rescale with the data, so it dominates and drives every column's precision to its maximum. **`effective_rank` reads 0 in every row**, including the rows where 13–74% of the variance is being removed: it is not a usable indicator of factor activity, and every claim in this paper about whether the factor engaged is made from the adjusted-vs-kept comparison instead.

multiplier	mean sd	removes	`eff. rank`	AUC
×0.01	0.038	**0.0%**	0	0.943
×0.1	0.375	**1.5%**	0	0.940
×1	3.752	74.1%	0	0.973
×3	11.256	72.5%	0	0.967
×10	37.521	71.1%	0	0.973
×100	375.207	65.4%	0	0.940
×1000	3752.070	12.6%	0	0.893

On CausalRivers the mechanism is mis-scaled, not merely idle. Over 25 flood/Random-5 subgraphs the column standard deviations inside a single five-station panel span 0.030 to 175.5 and *every* sample has at least one sub-unit column, so the floor bites on part of the panel and not the rest. Unrescaled (the published arm) the share removed is **negative on 14 of 25** samples — the adjustment *adds* idiosyncratic variance rather than removing a confounder. z-scored it is small but mostly positive (mean +0.041, negative on 4 of 25). **And the obvious objection is falsified**: residualisation does *not* remove the scale prior. Varsortability over the same subgraphs is 0.99 on the raw series and 0.76 on the CAFÉ residual — still far above the 0.5 no-signal point — with R^2-sortability 0.30 against 0.33. The ordering survives; our estimator is simply built not to read it (Table 5).

tween E=1 and E=2: −0.0101 at E=1 (winning on 1 of 8 seeds), then +0.0015, +0.0037, +0.0057, +0.0086. The mechanism is not subtle. The path and screening components are pooled across entities before they are ranked, so their variance falls like $1/E$ while their bias does not. At E=1 they carry the same bias with E times the variance, and rank-averaging them into the refit is strictly worse than the refit alone. **Every public single-series benchmark in this paper — CausalRivers, CausalDynamics, CauseMe — is** E=1, and `SCORE_WEIGHTS` was calibrated on a panel.

We report this as a *scope limit of the design*, and we deliberately did not ship a gate for it. The effect is 0.01–0.02 AUC; it changes no rank in Table 8, where we lose by 0.2–0.4 AUPRC. A weight schedule fitted to recover it would be exactly the benchmark-fitting this paper criticises in §4, bought for a hundredth of an AUC. The measurement is the deliverable.

Table 11: **The evidence blend needs cross-entity replication.** Same generator, same χ, same P=24; only the number of entities E varies. *Premium* is AUC(blend) − AUC(t_{direct}): positive means blending the three evidence statistics beats the strongest one alone. The FIXED-N block holds ET constant so the contrast is entities, not rows. Off-diagonal lagged AUC, mean over $\chi \in \{0.5, 0.9\}$ and seeds.

arm	$E \times T$	t_{direct}	path	screen	blend	premium	
fixed-N							
fixed-N	1 × 1200	0.856	0.725	0.722	0.846	-0.0101	(1/8)
fixed-N	2 × 600	0.898	0.824	0.812	0.900	0.0015 *	(5/8)
fixed-N	4 × 300	0.967	0.923	0.923	0.970	0.0037 *	(5/8)
fixed-N	10 × 120	0.963	0.934	0.939	0.969	0.0057 *	(7/8)
fixed-N	20 × 60	0.965	0.939	0.941	0.974	0.0086 *	(6/8)
fixed-T							
fixed-T	1 × 120	0.584	0.552	0.548	0.587	0.0028 *	(4/8)
fixed-T	2 × 120	0.682	0.629	0.635	0.703	0.0211 *	(8/8)
fixed-T	4 × 120	0.866	0.798	0.812	0.880	0.0137 *	(7/8)
fixed-T	20 × 120	0.989	0.968	0.975	0.992	0.0024 *	(5/8)

Two hypotheses we tested and falsified. Neither produced a change to the library, and both are reported because a diagnosis section that only lists confirmations is not a diagnosis.

(i) "Residualisation removes the signal where the common component is generated by the graph rather than confounding it." This has a sharp testable consequence. An exogenous factor with diagonal AR dynamics contributes a *symmetric* term $\Lambda\,\mathrm{diag}(a^k\sigma^2)\Lambda^\top$ to the lagged cross-covariance, while a directed edge contributes an *asymmetric* one — so removing a genuinely exogenous factor must *raise* the antisymmetric share of C_k. On 25 CausalRivers samples the share does rise, on all 25, exactly as predicted — and the AUC *falls* on 12 of them. Correlation between the two is +0.10 and sign agreement is 0.40, below chance. **Falsified.** The reason is measurable too: rivers discharge is near-unit-root, so $C_k \approx C_0$ and the antisymmetric share is 0.001–0.003. The statistic saturates and carries no information at that scale, against 0.10–0.46 on our own panels. A diagnostic that cannot resolve the case it was built for is not a diagnostic.

(ii) "The path component is uninformative when it cannot adjudicate comparisons." Measured resolution — $1 - \sum_v p_v^2$, the probability that two entries of the path statistic differ at all — runs 0.477 on our panels, where that component scores 0.957–0.991 AUC, and 0.67–0.87 on rivers, where it scores 0.398–0.750. Resolution is *anti*-correlated with informativeness, so the proposed gate would have fired in the wrong direction. **Falsified**, and no gate was built.

7 The point-in-time board

Every published time-series discovery method we could install is *batch*: it reads the whole series before emitting a graph, so a graph consumed at t — to build

a feature, gate a monitor, seed a policy — carries information from after t. Nobody reports this, so we measured it through one neutral standard: the same truncation-invariance certificate CAFÉ uses for imputation [2], with each method's own binary support as the indicator. The twelve baselines span constraint-based (PC, FCI [19]), constraint-based time-series (PCMCI [17], PCMCI+ [18]), continuous-optimisation (NOTEARS [13], DAGMA [14], DYNOTEARS [12]) and LiNGAM-family (DirectLiNGAM [15], VARLiNGAM [16]) methods; the board is now complete, eight tasks for every method including ours.

Two traps need handling. First, a method that asserts *no* edges passes the certificate vacuously; those rows carry a dagger, and FCI earns one on four of eight tasks. Second, the binary certificate is near-definitional for a batch method, so it is the least informative column on the board — CAFÉ-CD scores 0/8 revisions and 0.000 churn because the pipeline is truncation-invariant by construction, which is a statement about its design and not a contest it won. What is informative is the *magnitude*: churn, orientation flips, and the leakage delta Δ measured on each rival's own metric, own threshold and own data. VARLiNGAM's published AUPRC of 0.919 becomes 0.766 refitted on each prefix and read at the right edge — but its sample-size-matched delta is -0.009, which says that gap is almost entirely the *loss of data*, not look-ahead. DirectLiNGAM churns a quarter of its graph and flips 14 edges on average; DYNOTEARS is the most stable batch method here.

We are the ones who defined this test, so the honest framing is not that we win it. It is that this is a property of a discovery method nobody currently reports, it is cheap to report, and it costs different methods very different amounts.

8 Runtime, and proving the orientation

Runtime is the one place where the CPU-only, numpy-only design is unambiguously an advantage, and it is measurable without a leaderboard. On CauseMe's [7] only real-data experiment — `river-runoff`, N=12, T=4600, one series, one process, same CPU — **CAFÉ-CD takes** 2.42 s **against PCMCI+'s** 49.55 s**, a** 21× **gap**, at the identical fully dense 0.917 non-zero submission (Table 13a). The gap is smaller on toy sizes (3.6× at N=5, T=300) because the tigramite methods pay a subprocess spawn per dataset for GPL isolation; the N=12 number is the one to carry. On the panels of §3 CAFÉ-CD averages 4.01 s per cell (Table 2, last column), faster than every baseline there except VAR-OLS' 0.00 s — which is the honest other side of it.

Nothing was uploaded, and we say what that means. CauseMe scores server-side and its experiment zips ship no ground truth, so *no CauseMe leaderboard number appears in this paper*; 52 submission artefacts are built and validated against the platform's stated contract and left on disk. What *is* locally measurable is orientation, and it is the failure mode a server-scored benchmark cannot catch: a transposed submission does not score badly-but-plausibly, it scores inverted. We therefore write a known graph to disk in CauseMe's own format, read it back through the identical loader and adapter, and score it with CauseMe's own metric definitions. **All** 13 **methods beat their own transpose** (Table 13b); CAFÉ-CD scores 0.921 against 0.554 transposed. This is a format test, not a score.

9 Two engineering results

The engine router. On a panel, CAFÉ must decide whether one loading matrix can represent every entity's common component. The natural test — align the entities' principal subspaces and threshold — **fails**, because raw alignment is dominated by P, the rank and the sample size rather than by homogeneity: we measure raw between-entity alignments of 0.840–0.936 on homogeneous panels and 0.857–0.942 on heterogeneous ones, which interleave, so no absolute cutoff orders them. Dividing by each panel's *own* within-entity split-half alignment — the sampling-noise ceiling identical loadings would produce — gives a statistic whose null is 1 by construction, and it separates the families cleanly, picking the lower held-out-MAE engine on 8/8 panels (Table 4a). It is a property of the design matrix and never of an outcome, which is what keeps the routing causal-clean [2]. The scoring router of §6 is the same discipline applied to a different decision.

A null result on bit-packed screening. The lag screener's statistic is one bit per observation per scale, so a packed-bit popcount backend ought to beat a dense float matmul; that is the mechanism behind ARROW's reported speed-ups [20]. We implemented it and it lost, at every size tested, by 8–21×, with the gap *widening* in P and no crossover anywhere (Table 4b). The reason is that our baseline is not what ARROW accelerated: ARROW's headline speed-up is against a brute-force lag search dominated by histogram binning, whereas here the baseline is a BLAS-3 `sgemm` doing the same P^2n work at hundreds of GFLOP/s, and the popcount path is a memory-bound lookup-table gather that numpy must drive from a Python loop over targets. Beating `sgemm` would need a SWAR `uint64` or a JIT/Rust kernel, which a numpy-only module does not have. The backend is kept — scoring an arbitrary shortlist of (i, j, k) triples cannot use a matmul at all — but it is not the default, and the module documents the measurement rather than the aspiration. Separately: at the default 200 permutations the null calibration is 98% of the screener's wall clock. The statistic is nearly free; the calibration is the cost.

Table 12: **The point-in-time board: an unexamined property of time-series causal discovery, and what it costs each method.** Every published method here is *batch*, so a graph consumed at t carries information from after t. REV counts the tasks on which the method fails the truncation-invariance certificate CAFÉ uses for imputation (an edge asserted at a prefix is later rewritten); because a batch method fails that almost definitionally, the informative columns are the *magnitude* — CHURN, mean and max Jaccard distance between the graph asserted at a prefix and the graph finally published; FLIP, edges whose orientation reverses; $|E|$, final edge count — and the *leakage delta* Δ, on each method's own metric, threshold and data: published AUPRC minus right-edge honest AUPRC (refitted on each prefix, read at t). Δ_{match} repeats it at identical sample size, isolating look-ahead from the mere loss of data. Mean over 8 tasks (synthetic VAR, synthetic contemporaneous, CausalRivers, CausalDynamics; 2 each). † asserted *no* edges on ≥ 1 task, where the certificate is vacuous; FCI earns one on four of eight. REV prints its own denominator, so a row with fewer tasks (marked *) can never be read as a full-board mean. CAFÉ-CD's AUPRC columns here are a *lower bound*: this board's rows were computed before `CausalGraph.scores()` was changed from a two-tier step to continuous evidence. The certificate columns read the support and are unaffected. *We defined this test; it is offered as a property worth reporting, not as a ranking we win.*

	certificate					AUPRC			
Method	rev	churn	$\text{churn}_{\max}$	flips	$\lvert E\rvert$	pub.	honest	Δ	Δ_{match}
VARLiNGAM	8/8	0.142	0.357	2.5	8.9	0.919	0.766	+0.153	-0.009
DAGMA (linear)†	6/8	0.156	0.386	3.2	2.8	0.567	0.530	+0.036	-0.006
PC (lag-augmented)	6/8	0.107	0.229	3.4	5.9	0.634	0.597	+0.036	+0.012
VAR-OLS	8/8	0.095	0.239	0.6	11.9	0.778	0.759	+0.018	-0.014
PCMCI	7/8	0.146	0.303	1.2	11.1	0.787	0.770	+0.017	-0.005
DirectLiNGAM†	7/8	0.258	0.593	14.4	3.8	0.531	0.518	+0.013	-0.006
PCMCI+	8/8	0.166	0.336	6.8	6.6	0.671	0.659	+0.013	+0.016
Granger (pairwise)	6/8	0.115	0.229	5.0	11.9	0.740	0.730	+0.010	-0.009
FCI (lag-augmented)	6/8	0.109	0.228	0.5	5.1	0.629	0.623	+0.006	+0.009
DYNOTEARS	6/8	0.061	0.174	0.0	8.1	0.742	0.744	-0.002	-0.003
FCI†	4/8	0.201	0.500	0.1	0.4	0.424	0.433	-0.010	+0.004
PC†	5/8	0.317	0.625	6.8	1.6	0.400	0.423	-0.023	-0.001
CAFÉ-CD	0/8	0.000	0.000	0.0	13.5	0.626	0.608	+0.018	-0.006

10 Limitations

1. **We lose on both public single-series benchmarks.** Table 8 is the evidence and §6 is the diagnosis: an OLS VAR at ∼1 ms beats us in all 5 completed CausalRivers configurations, and residualising costs 0.278 AUROC there. A real gap remains once the defect below is accounted for, and it is not fixed today.
2. **CAFÉ's ARD rank selection is not scale-equivariant.** Table 10: the factor adjustment removes 74% of the variance at unit scale and *nothing* below it, because the +1 in $\alpha_l = N/(\|W_{\cdot l}\|^2 + 1)$ does not rescale with the data. On unrescaled CausalRivers it is misscaled rather than idle: the share removed is negative on 14 of 25 samples. The line is in the production imputation path and a fix needs the main CAFÉ benchmarks rerun to 10^{-12}; standardising inside `discover()` instead would make the scale prefix-dependent and could break the REV=0 certificate. Neither route is shipped.
3. **`effective_rank` does not report factor activity.** It reads 0 across the whole of Table 10, including rows removing 74% of the variance. Any claim about whether the factor engaged must come from the adjusted-versus-kept residual comparison.
4. **The evidence blend is calibrated for $E > 1$.** Table 11: the blend's premium over its strongest component is −0.0101 at E=1 and rises monotonically to +0.0086 at E=20. Every public single-series benchmark here is E=1. We did not gate on it — the effect is 0.01–0.02 AUC and changes no rank — but it is a known miscalibration.
5. **The head-to-head we win is a panel comparison.** In Table 2 the baselines are single-series methods run per entity. Pooling the entity dimension is a large part of the margin. The claim that transfers is "CAFÉ-CD is the right tool for panels with a pervasive confounding factor", not "CAFÉ-CD is a better discovery algorithm".
6. **Below the χ crossover the representation is worse than raw.** At χ=0.40 residualisation costs 0.059–0.072 AUC depending on the estimator.
7. **Most of the oracle headroom is unattainable.** At χ=0.95 the oracle residual scores 0.992 under OLS while the best estimated residual reaches 0.790. That gap is not slack better downstream estimation can close: it is the cost of *estimating* the factor, and it is larger than the confound removal itself.
8. **A linear model on chaotic ODEs is a structural mismatch.** The CausalDynamics systems are deterministic chaotic flows whose Jacobian sparsity is the truth; a lag-K linear model is misspecified for them by construction.
9. **CAFÉ-CD's AUPRC on the point-in-time board**

Table 13: **Runtime is a genuine win, and the orientation is proved rather than asserted. (a)** CauseMe `river-runoff_N-12_T-4600`: one *real* N=12, T=4600 series, one process, same CPU. CAFÉ-CD is 21× faster than PCMCI+ at the same fully dense submission (*nz*, the fraction of off-diagonal pairs given a non-zero score). **(b)** a synthetic round-trip through CauseMe's *own* on-disk format, loader and metric definitions (B=12, N=5, T=300): each method is scored against a known graph and against its own transpose. **All 13 beat their transpose**, so no submission is inverted — the failure mode that a benchmark with server-side scoring cannot catch. This is a format test, **not** a CauseMe score: the experiment zips ship no ground truth, so no CauseMe leaderboard number appears in this paper. 52 artefacts are built and validated; nothing was uploaded.

(a) s / series, N=12

method	s	nz
VAR-OLS	0.003	0.917
Granger-pw	0.15	0.917
PC	0.19	0.188
FCI	0.24	0.201
DirectLiNGAM	0.35	0.389
DAGMA	1.51	0.222
VARLiNGAM	1.85	0.778
CAFÉ-CD	**2.42**	0.917
PCMCI	8.31	0.917
PC-ts	10.66	0.146
DYNOTEARS	11.10	0.917
FCI-ts	12.41	0.153
PCMCI+	**49.55**	0.917

(b) orientation round-trip

method	AUC	⊤	margin
DYNOTEARS	1.000	0.551	+0.449
PCMCI	1.000	0.549	+0.451
VAR-OLS	1.000	0.565	+0.435
PCMCI+	1.000	0.514	+0.486
Granger-pw	0.990	0.590	+0.400
PC-ts	0.975	0.557	+0.418
FCI-ts	0.972	0.562	+0.410
VARLiNGAM	0.971	0.573	+0.398
CAFÉ-CD	0.921	0.554	+0.367
PC	0.609	0.486	+0.123
FCI	0.603	0.555	+0.048
DirectLiNGAM	0.554	0.531	+0.022
DAGMA	0.513	0.502	+0.011

is a lower bound. Those rows predate the change of `CausalGraph.scores()` from a two-tier step to continuous evidence. The certificate columns read the support and are unaffected; the CausalDynamics row of Table 8 was re-run after the change and the support reproduced on 54/54 systems, so the effect there was +0.001 AUPRC.

10. **Self-loop and threshold conventions differ across benchmarks and move scores materially.** CausalRivers removes the diagonal; CausalDynamics keeps it (a Jacobian diagonal is a real self-dynamics edge), which is what lets a constant identity matrix win there; CauseMe excludes self-links. Both benchmarks also report best-hyperparameter AUROC and oracle-threshold max-F_1; we match them for everyone, but they are upper bounds, not deployment numbers.
11. **Some head-to-head cells are still running.** Every affected cell is marked, listed in §11, and left empty rather than filled.

11 Reproduction and provenance

Every number here comes from a cache produced by a run in this repository or from a script in `scripts/` that we executed; no figure is illustrative and no cell is hand-transcribed. The build itself emits the table-to-source map and the list of cells left empty:

`gate.tex, figures/gate.pdf` ← `bench/causal_sparse_gate_cache.base.json`; `h2h.tex` ← `bench/causal_synthetic_h2h_cache.json`; `rivers.tex (sortability rows)` ← `data/varsortability.json (live)`; `rivers.tex, scaleprior.tex` ← `bench/bench_rivers_cache.json`; `dynamics.tex` ← `bench/bench_dynamics_cache.json`; `coupled.tex` ← `data/causal/cache/cd_coupled_audit.json`; `losses.tex` ← `bench/bench_rivers_cache.json + data/dynamics_matched.json (live)`; `diag.tex` ← `data/conditioning.json (live)`; `scale.tex` ← `data/scale_equivariance.json (live)`; `runtime.tex` ← `bench/bench_causeme_cache.json`; `pit.tex` ← `bench/pit_board_cache.json`; `router.tex` ← `data/engine_router.json (live)`; `screen.tex` ← `data/screen_backend.json (live)`; `sec. 6 text: the 40-sample residual bivariate re-measurement, the antisymmetric-share test and the path-resolution test` ← `measured in the scoring-router investigation (src/cafe/_skeleton.py, src/cafe/discovery.py); no cache in this directory`; `replication.tex` ← `bench/exp_causal_replication.py (bench/causal_replication_cache.json)`.

Left empty because the run had not finished at build time: CausalRivers free/east/confounder_3: the CAFÉ-CD arm has not finished (11 of 13 methods present), so no rank is printed; CausalDynamics coupled head-to-head: the two boards overlap on only 7 fully-scored systems (baseline board 10, CAFE-CD board 132), too few to rank, so no number is printed; CausalDynamics climate head-to-head: the two boards overlap on only 11 fully-scored systems (baseline board 12, CAFE-CD board 12), too few to rank, so no number is printed.

Rebuild with `scripts/make_tables.py` then `tectonic causal.tex`; the six live probes regenerate their own JSON first. Baseline installation, licence isolation (`tigramite` is GPL and runs out-of-process) and the self-test proving all 14 wrapper index conventions are in `docs/causal_baselines.md`.

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